

Pedestrians Movement on Crossings

Example Simulation Sessions

Course: Agent Systems

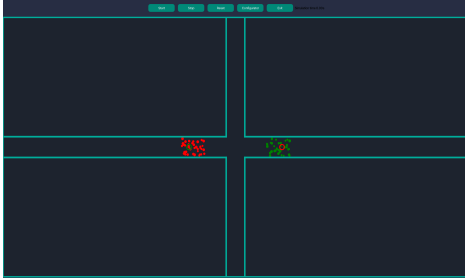
Authors:

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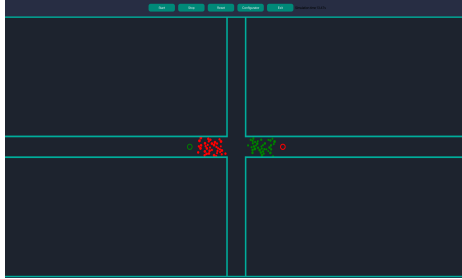
June 2026

Simulation 1 - Two-group bottleneck

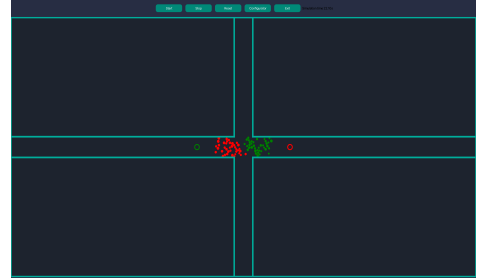
Two pedestrian groups (red, green) approach a central constriction from opposite sides and must pass through the same narrow opening in counter-flow. Frames are ordered in time; agent colour shifts from the group colour towards red as the local crowd pressure increases.



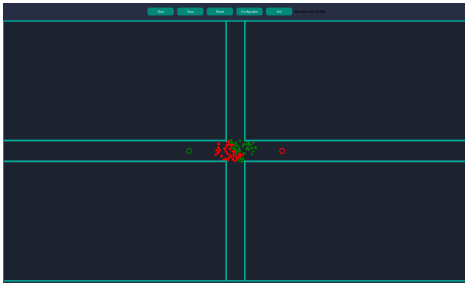
frame 1



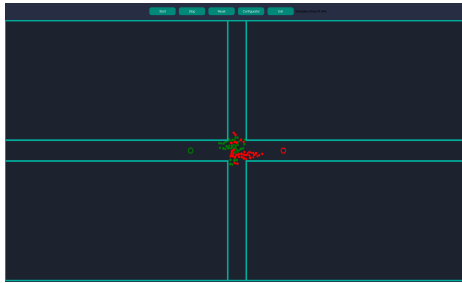
frame 2



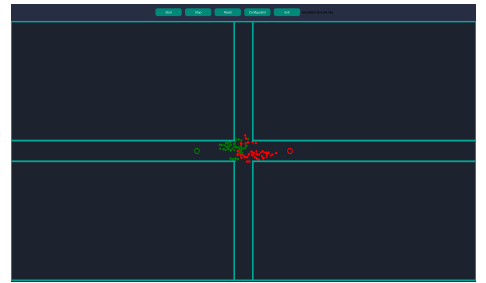
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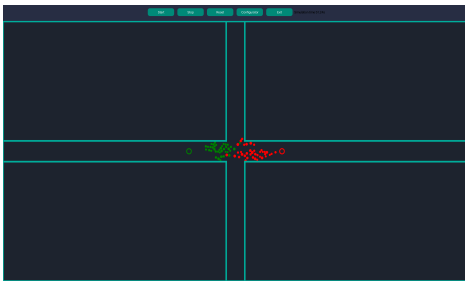
frame 4



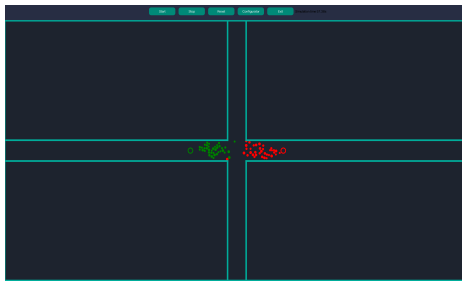
frame 5



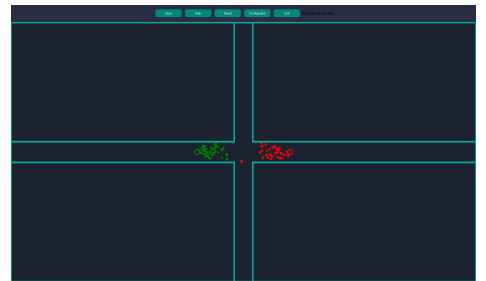
frame 6



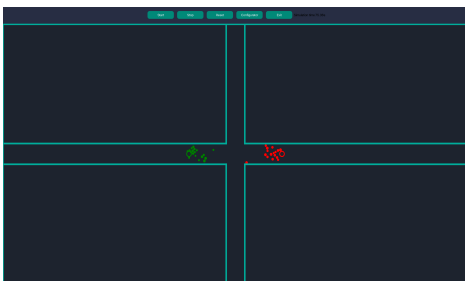
frame 7



frame 8



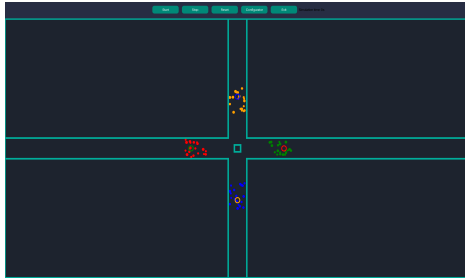
frame 9



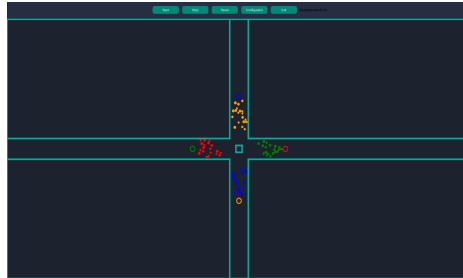
frame 10

Simulation 2 - Four-group crossing

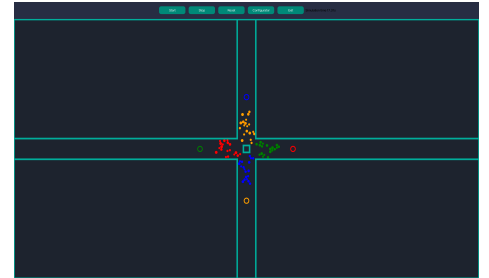
Four groups (red, green, blue, orange) enter an intersection from the four cardinal directions, each heading to the opposite exit. The groups intermix at the centre before separating again - note the pressure build-up where the four streams overlap.



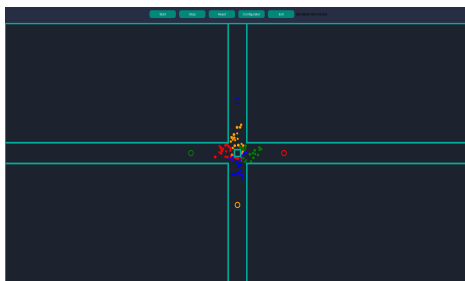
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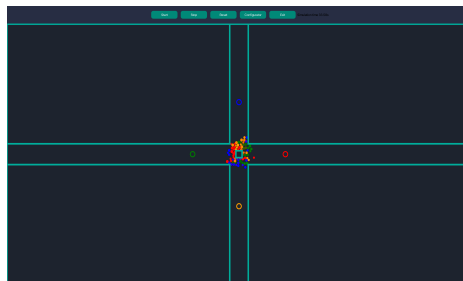
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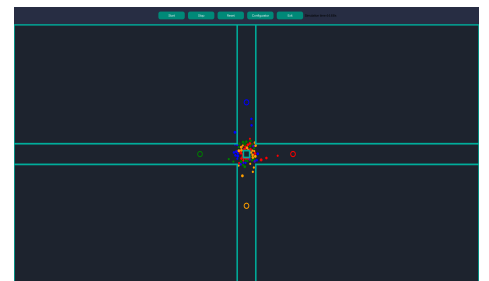
frame 3



frame 4



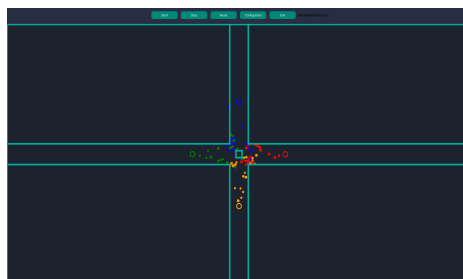
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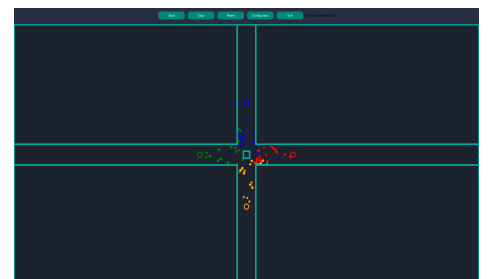
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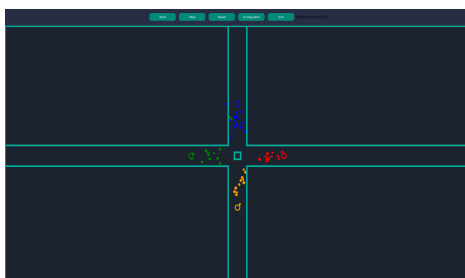
frame 7



frame 8



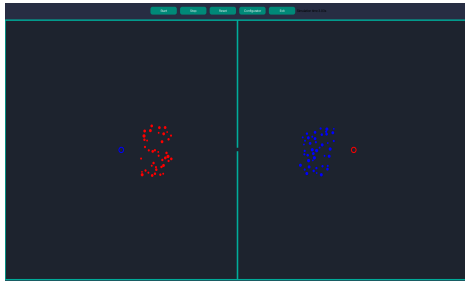
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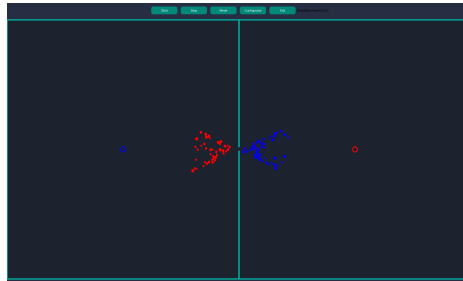
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Simulation 3 - Single narrow gap

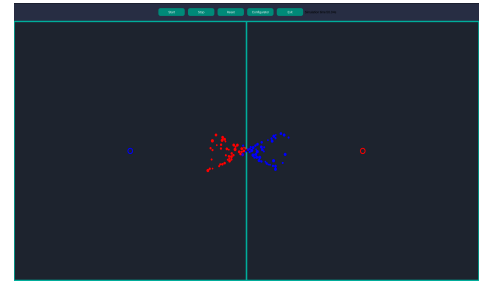
Two groups (red, blue) cross in opposite directions through a single narrow gap in a dividing wall. This is the most severe constriction and produces the longest clearance time and the highest sustained pressure at the opening.



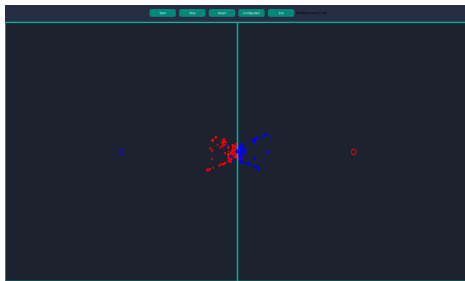
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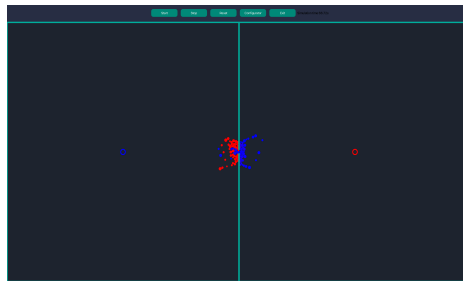
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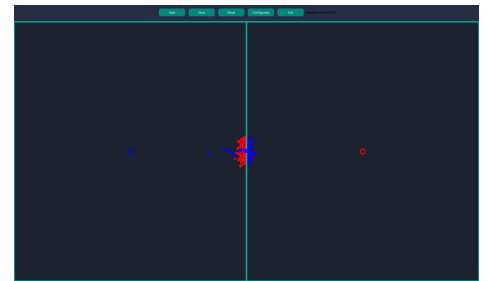
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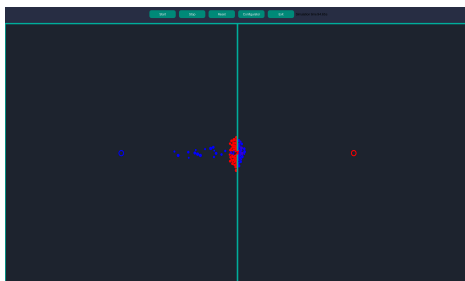
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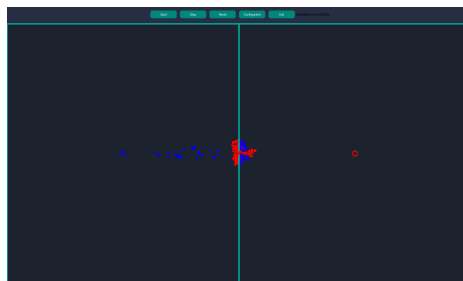
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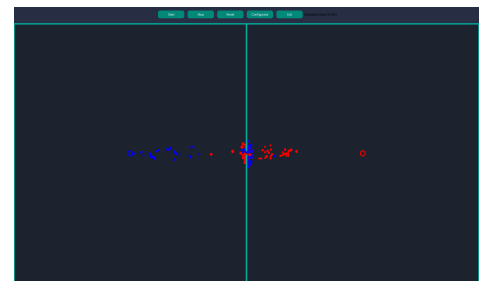
frame 6



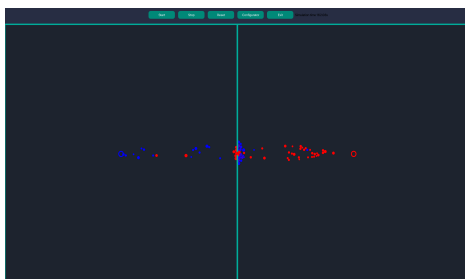
frame 7



frame 8



frame 9



frame 10